

Rayyan

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EDUCATION

University of Massachusetts Amherst

B.S. in Computer Science | GPA: 3.95/4.00 | Chancellor's Award, Dean's List

Coursework: Data Structures & Algorithms, Reasoning Under Uncertainty, Discrete Mathematics, Object-Oriented Programming, Linear Algebra, Multivariate Calculus

Amherst, MA

Expected May 2028

EXPERIENCE

Machine Learning Researcher

UMass Amherst | Python, PyTorch, MNE, scikit-learn | [Poster](#)

Jul 2026 – Sep. 2026

Amherst, MA / Remote

- Built a privacy evaluation pipeline using **PyTorch, MNE, and scikit-learn** to classify motor-imagery EEG signals from **9 participants** while measuring how much participant identity the data revealed.
- Compressed the neural network's learned EEG features from **64D to 8D**, reducing participant identification accuracy from **72.4% to 61.5%** while improving motor-imagery classification accuracy from **50.1% to 54.1%**.
- Tested a stochastic version of the **8D privacy model across 10 random seeds**, reducing participant identification accuracy from **61.5% to 39.2%** while retaining **48.3% motor-imagery accuracy**, compared with a **25% chance baseline**.

Software Engineer Intern

Myco | Web3 Video Streaming & Entertainment Platform | Python, LM Studio, FFmpeg, MLX Whisper

May. 2026 – Jul. 2026

Karachi, Pakistan

- Enriched **20,000+ videos across 43 metadata fields** by building a Python classification pipeline with LM Studio-hosted local LLMs and API models to identify content type, context, and genre.
- Turned unstructured video content into recommendation-ready metadata, supporting the development of Myco's system for suggesting more relevant videos to users.
- Doubled metadata-processing throughput by cutting processing time for a 20-minute video from approximately **4 minutes to 2 minutes**, consolidating FFmpeg, FFprobe, and MLX Whisper extraction and transcription into a unified Python pipeline.

Research Intern

Kandula Research Group | Python, TrackPy, NumPy, Pandas, ImageJ/Fiji

Oct. 2025 – May 2026

Amherst, MA

- Identified and characterized **4 recurring six-particle structures**, including a previously unobserved active-system pattern, by using NumPy and Pandas to build and analyze neighbor-based connectivity graphs.
- Reduced particle-tracking analysis time from **over 2 hours to approximately 5 minutes per dataset** by automating particle detection, tracking, and structural analysis across **30+ microscopy datasets totaling over 200 GB**.
- Validated classifications by using **Matplotlib** to overlay computed clusters onto raw microscopy frames, then ranking recurring structures across an average of **15 six-particle clusters per frame**.

Software Engineer Intern

Astrik Digital | Software Development Agency | React, TypeScript, Firebase, MongoDB

Jan. 2025 – Apr. 2025

Karachi, Pakistan

- Diagnosed client-facing API failures, broken forms, and CRM integration issues across **15+ websites** using browser DevTools, console logs, and network traces, delivering root-cause reports that restored lead-capture workflows.
- Modernized an internal food-ordering kiosk used by employees to browse menus and place workplace meal orders by rebuilding the C# application as a **React and TypeScript** single-page application integrated with **Firebase and MongoDB**.
- Slashed recurring crash and infinite-load incidents to near zero by replacing full-page reloads with client-side navigation and selective data loading, eliminating redundant network requests that failed on unstable in-store connections.

PROJECTS

Folio | Privacy-First, On-Device AI PDF Reader | [GitHub](#) | [Video Demo](#)

Personal Project

Electron, React, TypeScript, Node.js, PDF.js, llama.cpp, Tailwind CSS

- Architected a privacy-first Electron PDF assistant with a bundled **llama.cpp** server and typed IPC, keeping PDFs, annotations, and AI inference entirely on-device with no cloud API dependency.
- Delivered contextual PDF explanations in **under 4 seconds** by evaluating quantized Gemma models and configuring local inference with an **8,192-token context window**.
- Built a fault-tolerant model-management system with in-app GGUF downloads, streamed progress, load/unload controls, cancellation, and automatic cleanup of incomplete files, eliminating all terminal-based setup.

Closure | AI-Powered Workforce-Scheduling Platform | [GitHub](#) | [Video Demo](#)

1st Place, Hack(H)er413

React, Next.js, TypeScript, Supabase, Node.js, OpenRouter

- Achieved **~80% recommendation accuracy** by integrating OpenRouter to rank employees based on skills and availability, as measured against a fixed set of approximately 30 employee-shift scenarios with known correct assignments
- Shipped a responsive task-card interface that managed real-time state for tasks across employees, with priority badges, overdue indicators, and inline editing, persisting changes to Supabase.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript

Web & Full-Stack: React, Next.js, Node.js, Electron, Tailwind CSS, REST APIs, MongoDB, Supabase, Firebase

Machine Learning & Data: PyTorch, scikit-learn, MNE, Pandas, NumPy, Matplotlib, Hugging Face

Tools: Git, Linux, llama.cpp, OpenRouter, FFmpeg/FFprobe, MLX Whisper, TrackPy, ImageJ/Fiji, PDF.js